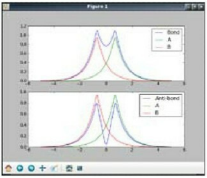

Figure 1: Bonding and anti-bonding 1s orbitals along the axis of a hydrogen molecule

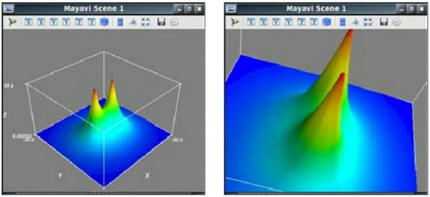

Figure 2: Two screenshots of a scene of the bonding 1s orbitals in the xy plane

```
from enthought.mayavi import mlab
mlab.surf(values, warp_scale='auto')
mlab.outline()
mlab.axes()
mlab.show()
```

The *mlab* module in Mayavi interfaces with the Scipy arrays. Create a 2D array using *mgrid*. The *surf* method plots the bonding function values against the x,y grid. The outline and axes are shown. Finally, the *show* method ensures that the image remains visible till you close the window. You can interact with the scene as illustrated by two images in Figure 2.

A molecule is a 3D object. So, what if you wanted to explore the orbital function in the x, y and z axes as well? One solution is to observe the contours:

```
import scipy as np
x,y,z = np.mgrid[-3:3:100j, -3:3:100j, -3:3:100j]
a = 0.7
a0 = 0.53
r1=np.sqrt((x-a)**2 + y**2 + z**2)
r2= np.sqrt((x+a)**2 + y**2 + z**2)
values = np.exp(-r1/a0)  + np.exp(-r2/a0)
from enthought.mayavi import mlab
mlab.contour3d(values,opacity=0.5)
mlab.outline()
mlab.axes()
mlab.show()
```

You have now created a 3D grid. The values of the electron density at each point is given in numbers. The method *contour3d* plots the contours, and you can interact with the image and view it from various perspectives. The opacity parameter is very important as it allows you to view the contours surrounded by other contours. One perspective is shown in Figure 3.

While contours are useful, you may wish to examine the values at each location. This is possible in Mayavi, as shown in the following example:

```
import numpy as np
x,y,z = np.mgrid[-3:3:100j, -3:3:100j, -3:3:100j]
a = 0.7
a0 = 0.53
r1=np.sqrt((x-a)**2 + y**2 + z**2)
r2= np.sqrt((x+a)**2 + y**2 + z**2)
values = np.exp(-r1/a0)  + np.exp(-r2/a0)
from enthought.mayavi import mlab
source = mlab.pipeline.scalar_field(values)
min = values.min()
max=values.max()
mlab.pipeline.volume(source,vmin=min,
vmax=min + .5*(max-min))
mlab.show()
```

Create a source for a scalar field and the volume method maps the source values to colours. The range of values to be mapped is given by the *vmin* and *vmax* parameters. A result can be seen in Figure 4.

The power of Mayavi/VTK is such that just a few lines of code enable you to create beautiful imagery, which can help get an insight into scientific data. The Scipy/Mayavi combination allows you to focus on the data and the observations.

You can explore more examples

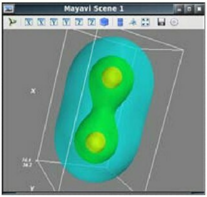

Figure 3: Contours of bonding 1s orbitals of a hydrogen molecule

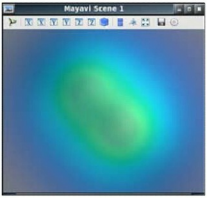

Figure 4: Volume rendering of the bonding 1s orbitals using partially transparent colours

of what is possible at *http://code.enthought.com/projects/mayavi/docs/development/html/mayavi/auto/examples.html.* END

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